

CUDA VS OpenCL

CUDA	OpenCL
Use compiler to build kernels	Build kernels at runtime
'C' language extensions—also a low-level driver-only API	API only; no new compiler
Buffer offsets allowed	API calls to execute kernel
Pointer traversal allowed	Buffer offsets not allowed

bridge between multiple disparate interfaces. With reprogrammable I/Os, these FPGAs are capable of supporting a wide variety of bridging, buffering and display applications. The recent emergence and rapid adoption of low-cost Mobile Industry Processor Interfaces (MIPI) such as CSI-2 and DSI has helped to simplify this task. By utilising the latest advances in I/O from the mobile computing market and MIPI together with the inherent advantages of low-density programmable logic in MHC architectures, designers can optimise their systems' ability to collect, transfer and analyse this key resource.

CUDA approach

CUDA architecture enables general-purpose computing on the GPU and retains traditional DirectX/OpenCL graphics. The dominance of multi-core systems in all domains of computing has opened the door to heterogeneous multi-processors. Processors of different compute characteristics can be combined to effectively boost the performance per watt of different application kernels.

GPUs and FPGAs are becoming popular in PC-based heterogeneous systems for speeding up compute-intensive kernels of scientific, imaging and simulation applications. GPUs can execute hundreds of concurrent threads, while FPGAs provide customised concurrency for highly parallel kernels. However, exploiting the parallelism available in these applications is currently not a push-button task. Often the programmer has to expose the application's fine and coarse grained parallelism by

using special application programming interfaces (APIs).

OpenMP

OpenMP (Open Multi-Processing) API supports multi-platform shared memory multiprocessing programming in C, C++ and Fortran. It consists of a set of compiler directives, library routines and environment variables that influence run-time behaviour.

OpenACC

OpenACC (for open accelerators) is a programming standard for parallel computing developed by Cray, CAPS, Nvidia and PGI. The standard is designed to simplify parallel programming of heterogeneous CPU/GPU systems. The programmer can annotate C, C++ and Fortran source code to identify areas that should be accelerated using compiler directives and additional functions.

The four steps to accelerate the code include: Identify parallelism, express parallelism, express data locality and optimise.

Selection of CUDA

The selection of CUDA as the programming interface for FPGA programming flow offers three main advantages:

1. It provides a high-level API for expressing coarse grained parallelism in a concise fashion within application kernels that are going to be executed on a massively parallel acceleration device.
2. It bridges the programmability gap between homogeneous and heterogeneous platforms by providing a common programming model for clusters with nodes that include GPUs and FPGAs. This simplifies application development and enables efficient evaluation of alternative kernel mappings onto the heterogeneous acceleration devices without time-consuming kernel code rewriting.
3. Wide adoption of the CUDA

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